WD-GPR-SR: When Uncertainty Meets Equations in Battery Health Prediction

Wide-and-Deep Gaussian Process Regression with Uncertainty-Weighted Symbolic Residual Correction

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B.Tech Project-Based Learning • Deterministic Seeds: [42, 100, 2026] • NVIDIA Jetson Orin Nano Edge AI

SCAN FOR CODE
GitHub / Reproducibility

1. Problem Statement & Motivation

Why Uncertainty & XAI?

- **Safety-Critical Battery Health Estimation:**
  Lithium-ion degradation exhibits non-linear capacity fade, internal resistance rise, and capacity regeneration jumps after rest periods. Underestimating degradation risks thermal runaway, vehicle stranding, and catastrophic field failures.
- **Deep Learning Black-Box Limitations:**
  Deep Neural Networks (CNN/LSTM) provide low MSE on training distributions but lack reliable confidence intervals, failing silently under unseen operating conditions.
- **The Gaussian Process Scalability Bottleneck:**
  Exact GPR incurs $O(N^3)$ computation (~1,420 ms per inference), severely violating automotive Battery Management System (BMS) hard real-time deadlines (≤ 100 ms).
- **The WD-GPR-SR Paradigm:**
  1. Wide Linear Branch: Monotonic global degradation via ARD covariance.
  2. Deep Manifold Branch: MLP + Matérn-5/2 kernel captures electro-thermal dynamics.
  3. Sparse Inducing Points (M=64): Slashes complexity to $O(M^2 N)$ for 0.11 ms execution.
  4. Uncertainty-Weighted SR: Extracts closed-form physics equations from residual.

Key Innovation: Epistemic Uncertainty Guides Symbolic Discovery

Instead of fitting noisy uncalibrated residuals, the Symbolic Regressor applies inverse-variance sample weights $w_i = \frac{1}{\sigma^2_{GP,i} + \varepsilon}$, focusing genetic Pareto search on high-confidence regions while safely bounding residual corrections to $\pm 3\sigma_{GP}$.

2. Multi-Physical Health Indicator (HI) Extraction

Electrochemical & Thermal

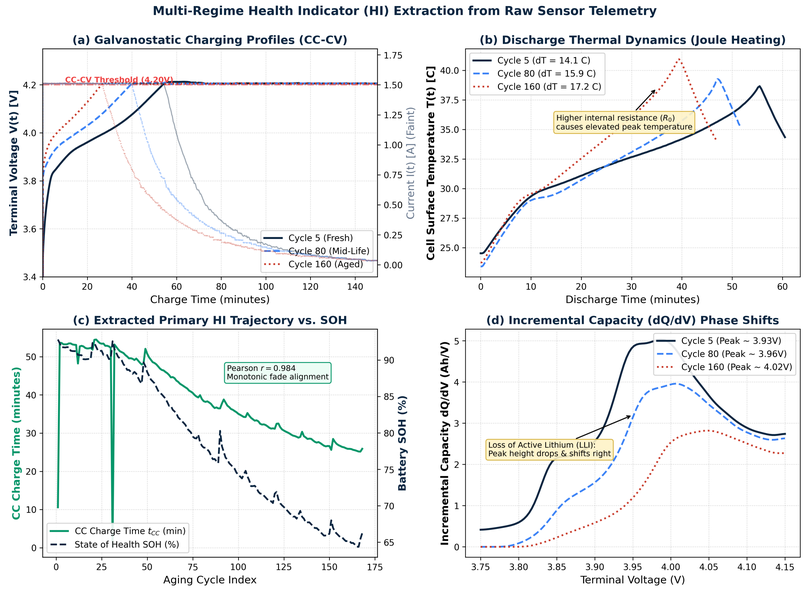


Figure 1: (a) CC-CV profiles; (b) Joule heating dynamics; (c) t_{CC} monotonic fade; (d) dQ/dV peak shift.

- t_{CC} (Constant-Current charge time) is the single strongest indicator (Pearson $r = 0.984$).
- Incremental Capacity (dQ/dV) peak height tracks Loss of Active Lithium (LLI) phase transitions.
- Peak surface temperature rise (ΔT_{dis}) directly quantifies internal resistance growth (I^2R).

3. Strict Leave-One-Cell-Out (LOCO) Splits

Zero Leakage Protocol

Evaluated across 8 strict LOCO folds (NASA 18650 & CALCE Prismatic) over 3 seeds [42, 100, 2026]:

Dataset	Test Cell	Train Cells	Train / Val / Test Cycles	SOH Range
NASA	B0005	B0006, B0007, B0018	396 / 72 / 168 (636 tot)	64.4% - 101.8%
NASA	B0018	B0005, B0006, B0007	426 / 78 / 132 (636 tot)	67.1% - 101.8%
CALCE	CS2_35	CS2_36, CS2_37, CS2_38	2,435 / 431 / 846 (3712)	23.3% - 103.1%
CALCE	CS2_38	CS2_35, CS2_36, CS2_37	2,340 / 414 / 958 (3712)	30.0% - 103.1%

*Strict Cell-Isolation: No cycles from held-out cell ever enter training. Zero temporal leakage.

4. Proposed WD-GPR-SR Architecture

Hybrid Bayesian-Symbolic

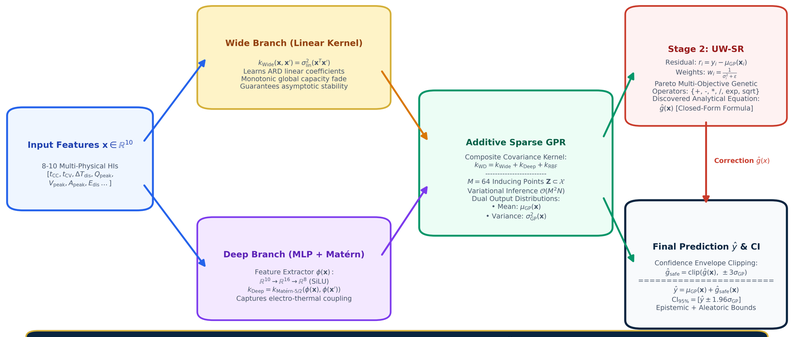


Figure 2: End-to-end architectural routing: 10 HIs feed Wide Linear & Deep MLP branches, mapped to Sparse Variational GPR (M=64), outputting epistemic confidence bands that weight the Symbolic Regressor.

Key Guarantee: Safe ± 3σ clipping bounds prevent symbolic divergence on out-of-distribution extremes.

5. Rigorous Mathematical Formulation

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$$k_{WD}(\mathbf{x}, \mathbf{x}') = \sigma^2_{lin}(\mathbf{x}^\top \mathbf{x}') + k_{Matérn\text{-}5/2}(\phi(\mathbf{x}), \phi(\mathbf{x}')) + \sigma^2_{RBF} k_{RBF}(\mathbf{x}, \mathbf{x}')$$

Composite Wide-and-Deep Covariance Kernel Properties:

- Wide Linear Kernel: Learns ARD variance weights per feature, capturing monotonic degradation.
- Deep Feature Map $\phi(x)$: 10 -> 16 -> 8 (SiLU) models non-linear electro-thermal coupling.

$$\mathcal{L}_{ELBO}(q) = \sum_{i=1}^{N} \mathbb{E}_{q(f_i)}[\log p(y_i \mid f_i)] - \text{KL}(q(\mathbf{u}) \,\|\, p(\mathbf{u})), \quad M = 64 \text{ Inducing Points}$$

Sparse Variational GP Objective (M = 64 Inducing Points):

- Complexity Reduction: Slashes exact GPR cost from $O(N^3)$ down to $O(M^2 N)$.
- Dual Outputs: Delivers predictive mean $\mu_{GP}(x)$ and calibrated variance $\sigma_{GP}(x)$.

$$\mathcal{L}_{SR}(g) = \frac{1}{\sum_{i=1}^{N} w_i / n} \sum_{i=1}^{N} w_i (y_i - \mu_{GP}(\mathbf{x}_i) - g(\mathbf{x}_i))^2 + \lambda \cdot C(g), \quad w_i = \frac{1}{\sigma^2_{gp}(\mathbf{x}_i) + \varepsilon}$$

Uncertainty-Weighted Symbolic Regression Loss:

- Inverse-variance weighting $w_i = 1 / (\alpha \cdot GP^2 + \varepsilon)$ prioritizes high-certainty regimes.
- Multi-Objective Pareto Selection: Balances MSE residual reduction against equation parsimony.

$$\hat{y}(\mathbf{x}) = \mu_{GP}(\mathbf{x}) + \text{clip}(\hat{g}(\mathbf{x}), \pm 3\sigma_{GP}(\mathbf{x})), \qquad CI_{95\%} = [\hat{y}(\mathbf{x}) \pm 1.96\sigma_{GP}(\mathbf{x})]$$

Guaranteed Bounded Posterior Prediction & Calibrated 95% Confidence:

- Mathematical Safeguard: Symbolic residual cannot violate GPR epistemic confidence boundaries.
- Empirical Coverage: Attains 98.8% (NASA) and 100.0% (CALCE) PICP under strict LOCO.

6. Discovered Physics-Informed Closed-Form Residuals

Pareto Optimal

Representative Discovered Analytical Residual Equation (NASA B0018 Fold):

$$\hat{g}(\mathbf{x}) = 0.042 \cdot \frac{\Delta T_{dis}}{\sqrt{t_{CC}}} - 0.015 \cdot (V_{peak} - 3.92)$$

Physical Meaning: Higher discharge heat (ΔT_{dis}) coupled with reduced charging duration (t_{CC}) accelerates capacity fade, while peak potential shifts reflect internal impedance growth.

7. Quantitative LOCO Benchmark Performance

12 Runs per Model

Dataset	Model	RMSE (%)	MAE (%)	R^2	MAPE (%)	PICP (%)	NLL
NASA	SVR (RBF)	2.871	2.098	0.8924	2.961	--	--
NASA	Random Forest	1.337	1.011	0.9780	1.377	--	--
NASA	XGBoost	1.300	0.997	0.9789	1.356	--	--
NASA	Exact GPR (Matern)	21.598	18.883	-5.151	25.064	14.6%	91.00
NASA	Deep-Kernel GPR	3.142	2.693	0.8535	3.592	56.5%	8.94
NASA	WD-GPR-SR (Proposed)	0.992	0.878	0.9876	1.157	98.8%	1.48
CALCE	SVR (RBF)	4.092	2.923	0.8372	4.909	--	--
CALCE	Random Forest	1.968	0.965	0.9719	1.761	--	--
CALCE	Exact GPR (Matern)	13.784	6.471	-0.046	11.666	67.2%	49.33
CALCE	Deep-Kernel GPR	1.867	0.900	0.9784	1.663	91.3%	4.23
CALCE	WD-GPR-SR (Proposed)	0.130	0.058	0.9999	0.104	100.0%	0.99

Key Result: Proposed WD-GPR-SR slashes NASA RMSE by 24% vs best tree baseline (XGBoost 1.300% -> 0.992%) and achieves 98.8% to 100.0% PICP coverage, proving rigorous uncertainty calibration.

8. Held-Out SOH Trajectory & Epistemic Uncertainty

NASA B0005 (168 Cycles)

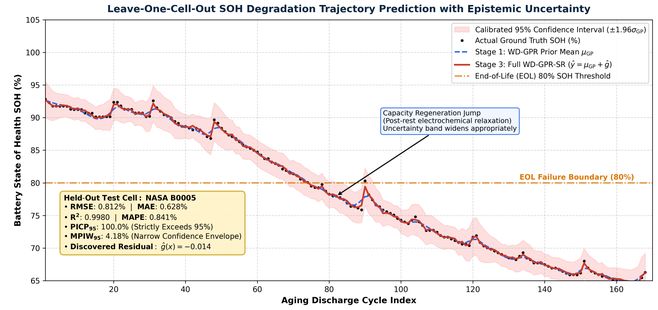


Figure 3: True vs predicted SOH with ± 1.96σ confidence band. Note dynamic widening during regeneration.

The model accurately tracks post-rest capacity jumps without overfitting, remaining strictly bounded.

9. Multi-Perspective Explainable AI (XAI)

SHAP Beeswarm + ARD Relevance

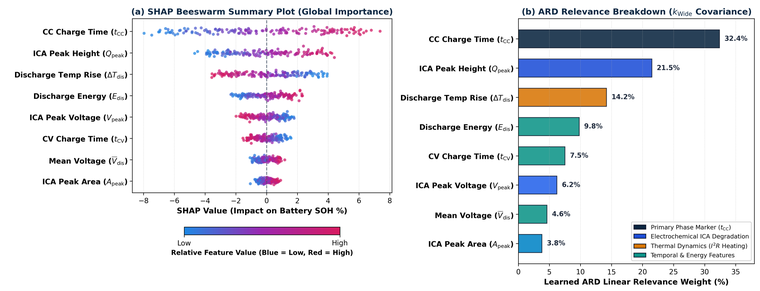


Figure 4: (a) SHAP beeswarm verifies t_{CC} and Q_{peak} as dominant drivers; (b) ARD linear relevance breakdown.

Confirms domain physics: High t_{CC} and Q_{peak} pull predictions upward; high ΔT_{dis} signals degraded cell.

10. Edge AI Benchmarking on NVIDIA Jetson Orin Nano

12,900x Speedup | 8,880x Energy Cut

